

Capstone 2: Business Analysis with Python

For this project, you will be working through reading, cleaning, and analyzing data for a fictional business using sample data. This project will call on your ability to read data and extract useful information using your Python skills.

Requirements

Your capstone project must include all of the following to be considered complete:

- A Jupyter Notebook that follows the setup outlined in Steps 1 through 5 and then contains your full Core Marketing Analysis, Steps 6 through 8.
- Included in your notebook, at least two Matplotlib visualizations illustrating your analysis
- All material published in your Capstone_2 repository by 8:30am Friday, with at least 1 commit per day to demonstrate your progress throughout the week
- A short README file in your Capstone_2 repository to describe the project
- 10-minute video recording to demonstrate your project and results, with a link to the recording in your Capstone_2 README file
- 5-minute live in-class presentation of sales territory analysis

You will also participate in peer review of your classmates' projects including review of their recordings and project files. If you have additional time during the week, you may also respond to the bonus questions included in the Appendix.

Note: You will have plenty of opportunities in this project to keep you busy and stretch your skills. Before you go overboard with diving into the bonus questions, make sure the core requirements are met!

Scenario

EmporiUm is a growing student bookstore chain that also sells tech gear, art supplies, and other products both in-store and online. Sales are segmented by region, with each region including multiple sales territories. The retail marketing team for each sales segment is responsible for marketing strategy to get customers in the door and buying products.

A marketing manager has asked you for some analysis comparing sales in the two territories assigned to you. This marketing manager happens to have a background in data analytics, so they have asked for your full analysis to be presented in the form of a Jupyter Notebook that walks through each step of the process and the results that you find, including at least two charts to visualize those results. Your video demo and live presentation should share the results of this analysis.

Additionally if time permits, your team manager has sent you a number of other questions to investigate and respond to using the data provided (in the Appendix). This can optionally be included in your video demo as well.

Project Setup

1. Start by creating a new repository named `Capstone_2`. Include a `README` file that provides a brief description of your project, including what sales territory you will be analyzing. Then use Git Bash to clone your new `Capstone_2` repo to your local directory `Documents/DATA_Labs`.
2. Download the sample data provided to your `Capstone_2` directory, then review it.
 - * Take some time to familiarize yourself with what's there. What files are included? What information is present in each? What are the pieces of information shared across files that will enable you to merge data for analysis?
3. Create a new notebook named `lastname_sales_analysis.ipynb` and save it to `Capstone_2`.
 - * Include a markdown cell at the top to describe the project and identify which sales territories you are analyzing.
4. Import the data analysis libraries you need for this project:
 - * One at a time, load each CSV file provided in the project material using `pd.read_csv()`, with a new code cell for each file imported. After import, confirm the column names, number of non-null records, and dtype for each column using the `.info()` method.
 - * Optionally, you might want to preview each DataFrame using the `.head()` method. (Do *not* display the full contents of every DataFrame! That leads to a lot of unnecessary scrolling.)
5. Add a markdown cell with the header **Core Marketing Analysis**.

Core Project Requirements

6. The marketing manager wants to know:
 - * Who are the territory managers for the sales territories assigned? What are the store IDs and cities for the stores in each assigned sales territory?
 - * What is monthly total revenue for in-store sales in each of the two sales territories, over the full period covered by the data?
 - * How would you rank the sales performance of each store in each sales territory? Which are the top-performing stores?
 - * Comparing the customer ID from the customer list data with the rewards ID from the sales data, who were the top customers in each sales territory?

- * What is the number of transactions per month by product category in each assigned territory? What is total sales revenue per month by category? What might this tell you about the most popular products, and where could there be opportunity for growth?
 - * What is your recommendation for where to focus marketing attention in the next quarter?
7. For each question above, write a series of markdown and code blocks to break down the question into parts and perform an analysis of the data to find an answer.
- * For the final question seeking a recommendation of where to focus marketing attention in the next quarter, use what you have discovered so far with the rest of your analysis (and optionally any additional analysis that you would like to explore), and add a markdown cell explaining what your recommendation is and why.
 - * Make sure to save your work and commit changes as you go!
8. Pick two or more of the questions that you analyzed above to render as a chart using Matplotlib.
- * Use appropriate chart types based on whether you are looking at trend over time, comparison across categories, or other type of analysis. Include clear labeling on the chart to describe what it depicts.
 - * Note: Put your charts wherever feels most logical in your notebook. You do not have to squash them down at the end (although you can put them at the end if that makes sense for your presentation).
9. Test your code frequently as you work, and periodically use Run All to check that the order of steps works across the full project. Be prepared to run your notebook live during your presentation; the notebook should execute cleanly from top to bottom when you restart the kernel and select Run All.

- Capstone Video -

10. Record a 10-minute video to present your capstone project. In your video:
- * Describe the steps you took to perform your analysis and what you found.
 - * Walk through each question one at a time. For your final recommendation, describe how you reached that recommendation.
 - * For your charts, explain what that chart depicts and why you chose that chart type. Point out any customizations you made to the chart.
 - * If you tackled any of the bonus questions, you may optionally include those in your video. (Keep your video to 10 minutes. If you don't have time to include bonus work, don't worry -- you will still get credit for that work if it is published to your Capstone_2 repository.)

- Bonus Questions -

11. If time permits: Create notebook with markdown and code blocks addressing each of the questions attached in the Appendix. Name the script `lastname_bonus_analysis.ipynb` and save it to your `Capstone_2` repository.
12. If time permits: Pick one or more of the questions you answered from the Appendix and create a chart to illustrate your answer. Include your chart(s) in `lastname_bonus_analysis.ipynb`.

- In-Class Presentations -

13. You will be delivering a short presentation of your sales territory analysis to your classmates. This is an important “data storytelling” opportunity! To get prepared:
 - * Practice delivering your presentation in advance. Rehearse what you will share on your screen and the words that you plan to say, and make notes to help yourself stay on-track.
 - * Make sure your project files are well organized so that you can find things quickly and smoothly.
14. When it is your turn to present, imagine you are meeting with the marketing team managers.
 - * **You will have 5 minutes for your presentation.** Be prepared to walk through the results of your analysis, the charts you created, and your recommendation for the next quarter.
 - * Share your screen while presenting. Use notes as needed to help you walk through your presentation in a well-organized manner.
 - * **NOTE:** Your Jupyter Notebook is the visual presentation of your report, and it should support the “data story” you want to convey. The most important thing is showing your understanding of the data and the processes that you have used to analyze it!
15. **When your classmates are presenting, give your full attention. You are encouraged to use Zoom reactions and chat during your classmates’ presentations to react to things you find helpful, interesting, or insightful.**
 - * **Consider also taking notes on the projects where you have been assigned to provide peer review (see below).** You should think about at least one question you have and at least one piece of feedback to share.

- Capstone Peer Review -

16. On presentation day, during your independent work time, you will be assigned 4 of your classmates’ video projects to watch and provide a peer review via Teams using the questions provided in your Week 8 course material.

- * Peer reviews should focus on **constructive feedback**. The purpose of providing feedback is not merely to state your opinion; rather, it is to help the other person improve and refine their work.

Tips for a successful capstone project

- Make sure you implement all the requirements from page 1!
- Save all your project files to your Capstone_2 repository. Make sure you regularly commit changes using meaningful commit messages. Ideally you should have multiple commits every day during capstone week.
- Use good naming conventions throughout your project (e.g. clearly labeled file names, appropriate use of snake_case and PascalCase in your code, etc). Add explanatory comments to your code throughout. This will make it easier for you to keep track of what you are doing, and also easier for your reviewers to understand your work.
- On presentation day, have the files you plan to share open and ready to share, so that you're prepared to start when your turn comes up.
- Make sure that the substance of your project reflects *your own independent work*. This is your opportunity to test your skills and show off what you can do!

APPENDIX: Capstone Bonus Questions

Create a script named `lastname_bonus_analysis.ipynb` and save it to your `Capstone_2` repository. Copy and paste the below questions into your notebook, each in a separate markdown cell. Below each question, provide Python code blocks to produce your answer. Add markdown blocks or code comments as needed to explain your answer.

Filtering, Sorting & Exploration

1. Create a list of all transactions from January 2024, sorted by sale amount from highest to lowest. Which transactions had a sale amount greater than \$500?
2. Find all products whose product number begins with 10525. What category and subcategory do they belong to?
3. Which stores in your assigned territories had at least one transaction with a sale amount over \$1,000? How many such transactions did each store have?

Aggregation & NumPy

4. What is the total sales revenue across all in-store transactions in the dataset? What is the average transaction amount? What is the standard deviation?
5. Which stores had total revenue above the average store revenue across the full dataset? (Hint: use `np.mean()`.)
6. What is the total revenue for each product category across all territories, sorted from highest to lowest? Which category has the most individual transactions?
7. Across the full dataset, what are the top 10 best-selling individual products by total revenue?

Merging & Grouping

8. For each transaction in your territories, display the transaction date, sale amount, city, state, and territory manager. (Requires merging `StoreSales` with `StoreDetail`.)
9. What is total revenue by region across the full dataset? Which region generates the most revenue? (Requires merging with `StoreDetail` and grouping by Region.)

10. For each product subcategory in your territories, calculate the total revenue and number of transactions. Which subcategories are driving the most revenue? (Requires merging StoreSales → Products → ProductCategories.)
11. Using the customer list, identify all rewards members who made purchases in your territories. What is the average transaction value for rewards members compared to non-rewards customers? (Requires merging StoreSales with customer_list on RewardsID = cust_id.)
12. Who are the top 10 rewards members by total spend across the full dataset? How many transactions does each have?

Advanced / Stretch

13. Compare year-over-year revenue for your territories. Is each territory trending up or down between years?
14. Identify the single best month and single worst month for each of your territories over the full data period.
15. Using your monthly revenue totals, fit a linear regression model to the data and plot the trend line alongside the monthly revenue chart. Is your territory trending up or down overall?